

遥操作使用说明

上位机控制程序demo

 follower_demo.zip

v2.5.7xml示例  kaanh.xml

配置

1. 新增规划器，设置要控制的关节和末端

```
<kaanh.multi_motion.MultiModelControl
name="kaanh::multi_motion::MultiModelControl" is_use_sp="false"
is_check_solve="true">
  <kaanh.AccDecJerkLimit __prop_name__="acc_dec_jerk_limit" joint_jerk="10"
cart_line_acc="2.5" cart_line_jerk="10" cart_angle_acc="2.5" cart_angle_jerk="10"
/>
  <kaanh.SpeedLimit __prop_name__="speed_limit" v_tcp_max="6" v_tcp_min="0"
w_tcp_max="12.56637061435917246" w_tcp_min="0"
max_manual_line_speed="0.20000000000000001" max_manual_angle_speed_per="30" />
  <kaanh.ZoneLimit __prop_name__="zone_limit" dis_max="0.10000000000000001"
dis_min="0" per_max="0.5" per_min="0" />
  <kaanh.CouplingConfig __prop_name__="coupling_config" is_coupling="false"
type="ToolTool" />
  <TgConfigVec __prop_name__="tg_config_vec">
    <!-- 多模型下如果需要控制部分末端，在这里新增配置 -->
    <!-- 如果要控制所有末端，不需要修改 -->
    <kaanh.TgConfig tg_id="0" is_cart="false" ee_ids="{0,1,2,3,4,5}" />
    <kaanh.TgConfig tg_id="1" is_cart="true" ee_ids="{0}" />
  </TgConfigVec>
</kaanh.multi_motion.MultiModelControl>
```

2. 跟随模块配置笛卡尔和关节跟随的tg_id号以及通讯设置

```
<!-- 使用socket通讯 支持TCP UDP WEB三种方式 -->
<UDPFollower name="UDPFollower">
  <FollowerParamPool __prop_name__="follower_param_pool">
    <FollowerAjParam tg_id="0">
      <SocketServer __prop_name__="socket_server" connect_type="UDP"
port="9999" />
    </FollowerAjParam>
    <FollowerCartParam tg_id="1">
      <SocketServer __prop_name__="socket_server" connect_type="UDP"
port="9998" />
    </FollowerCartParam>
  </FollowerParamPool>
</UDPFollower>
```

```
</FollowerCartParam>
</FollowerParamPool>
</UDPFollower>
```

状态获取

- 前端下发get指令
- 返回值中包含
 - ret_code
 - ret_msg
 - ret_context
- ret_context中"follower_msg"节点为跟随模块返回值，具体包括
 - running_state：当前是否在跟随
 - follower_mode：当前跟随的模式：笛卡尔&关节

get返回示例

[复制代码](#)

```
{
  "ret_code": 0,
  "ret_context": {
    "area_msg": {
      "states": [
        {
          "active": false,
          "captured": false,
          "control": false,
          "type": "None"
        },
        {
          "active": false,
          "captured": false,
          "control": false,
          "type": "None"
        },
        {
          "active": false,
          "captured": false,
          "control": false,
          "type": "None"
        },
        {
          "active": false,
          "captured": false,
          "control": false,
          "type": "None"
        },
        {
          "active": false,
          "captured": false,
          "control": false,
          "type": "None"
        },
        {
          "active": false,
          "captured": false,
          "control": false,
          "type": "None"
        },
        {
          "active": false,
          "captured": false,
          "control": false,
          "type": "None"
        },
        {
          "active": false,
          "captured": false,
          "control": false,
          "type": "None"
        }
      ],
      "conveyor_msg": {
        "catch_size": 0,
        "miss_que_size": [0],
        "miss_size": 0,
        "net_state": [false],
        "photo_size": 0,
        "pos": [0.0],
        "que_size": [0],
        "state": [false],
        "vel": [0.0]
      },
      "current_msg": {
        "collision_detection_sensitivity": [10.0, 10.0, 10.0, 10.0, 10.0, 10.0],
        "collision_detection_state": [0.0],
        "collision_detection_switch": false,
        "cur_status": false,
        "driver_data_msg": {
          "value": [
            [262142.0, 0.0, 0.0, 0.0],
            [262140.99999999997, 0.0, 0.0, 0.0],
            [262145.0, -0.0, 0.0, 0.0],
            [262140.0, 0.0, 0.0, 0.0],
            [262144.0, 0.0, 0.0, 0.0],
            [262145.0, 0.0, 0.0, 0.0]
          ],
          "value_info": [
            [
              [262142.0, 262142.0, 262142.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0]
            ],
            [
              [262140.99999999997, 262140.99999999997, 262140.99999999997],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0]
            ],
            [
              [262145.0, 262145.0, 262145.0],
              [-0.0, -0.0, -0.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0]
            ],
            [
              [262140.0, 262140.0, 262140.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0]
            ],
            [
              [262144.0, 262144.0, 262144.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0]
            ],
            [
              [262145.0, 262145.0, 262145.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0],
              [0.0, 0.0, 0.0]
            ]
          ]
        },
        "driver_msg": {
          "driver_break_ready": [true, true, true, true, true, true],
          "driver_err_code": [0, 0, 0, 0, 0, 0],
          "external_auto_msg": {
            "servoConfirm_shield": false,
            "follower_msg": {
              "follower_mode": "None",
              "running_state": false,
              "force_msg": {
                "fc_status":
```

```
[[],\\"fs_binded_ee\\":[],\\"fs_car_data\\":[],\\"fs_protection_switch\\":  
[],\\"fs_raw_data\\":[],\\"fs_switch\\":[],\\"fsg_status\\":false,\\"fsl_id\\":  
[]},\\"io_msg\\":{\\"digital\\":[\\\\"Actual\\",  
[false,false,false,false,false,false,false,false,false,false,false,false,false,  
e,false,false],  
[false,false,false,false,false,false,false,false,false,false,false,false,false,fals  
e,false,false]]},\\"log_msg\\":  
{\\"has_error\\":false,\\"id\\":4,\\"state_id\\":0},\\"motion_msg\\":{\\"actual_pos\\":  
[[0.0,0.0,0.0,0.0,0.0,0.0]],\\"actual_toq\\":  
[[0.0,0.0,0.0,0.0,0.0,0.0]],\\"jog_coordinate\\":\\"Axis\\",\\"jog_coupling_ids\\":  
[0],\\"jog_coupling_mode\\":\\"JogEe\\",\\"jog_mode\\":  
{\\"run_mode\\":\\"Continuous\\",\\"step\\":[0.0,0.0]},\\"jog_vel\\":10,\\"motion_state\\":  
[true,true,true,true,true,true],\\"motor_pos\\":  
[[0.0,0.0,0.0,0.0,0.0,0.0]],\\"part_pq\\":[[[0.0,0.0,0.0,0.0,0.0,0.0,1.0],  
[0.0,0.0,0.0,0.0,0.0,-0.0,1.0],[0.0,0.0,0.0,0.0,0.0,-0.0,1.0],  
[0.0,0.0,0.0,0.0,0.0,-0.0,1.0],[0.0,0.0,0.0,0.0,0.0,-0.0,1.0],  
[0.0,0.0,0.0,0.0,0.0,-0.0,1.0],[-5.8031691057605634e-12,-5.218048215738236e-  
15,6.661338147750939e-13,-1.745740669421564e-15,1.7457406694215694e-  
15,-1.6155445744325899e-15,1.0]]],\\"pe\\":  
[[0.0,181.10000000000002,1494.5,44.99999999999991,-89.99999999999972,224.999999999  
999]],\\"pe_config\\":[[0,0,0,-1]],\\"pe_vel\\":[0.0,0.0],\\"pq\\":  
[[0.0,181.10000000000002,1494.5,-0.5000000000000027,-0.4999999999999991,-0.49999999  
99999991,0.4999999999999991]]},\\"pallet_msg\\":{\\"count\\":[1],\\"serial_num\\":  
[0]},\\"program_msg\\":{\\"is_edited\\":false,\\"last_stopped_motor_pos\\":  
[],\\"loaded_info\\":[\\\\"\\",\\\\"\\",\\\\"loop_stack\\":[],\\"mode\\":\\"Continuous\\",\\"pc\\":  
[\\\\"\\",0],\\"run_direction\\":\\"Forward\\",\\"state\\":\\"Unloaded\\",\\"stmt_state\\":\\"None  
\\",\\"vel\\":50},\\"protocol_msg\\":{\\"mtcp_slave\\":  
{\\"has_connected\\":false,\\"is_serving\\":true}},\\"robot_msg\\":  
{\\"activate\\":\\"Disabled\\",\\"cs_error_code\\":0,\\"cs_error_msg\\":\\\\"\\",\\"motion\\":\\"S  
topped\\",\\"op_mode\\":\\"Manual\\",\\"status\\":\\"Normal\\",\\"tool\\":[0],\\"tool_mode\\":  
[\\\\"Tool\\",\\\\"wobj\\":[0]],\\"splc_msg\\":  
{\\"selected_prj\\":\\\\"\\",\\"state\\":\\"Unloaded\\",\\"system_sema_msg\\":  
{\\"gate_shield\\":false,\\"modbus_disconnect_shield\\":true,\\"raster_shield\\":false},\\"  
user_management_msg\\":  
{\\"user\\":\\"Engineer\\",\\"user_slave_msg\\":null},\\"next_msg\\":\\""}}
```

跟随控制指令

1. start_follower: 开始跟随功能
2. stop_follower: 关闭跟随功能
3. follower_aj: 运行轴关节跟随指令
4. follower_cart: 运行笛卡尔跟随指令

笛卡尔

1. 跟随方式：跟随最新位置
2. 跟随速度：
 - a. 最大跟随速度：点动笛卡尔设置的速度

b. 调速：最大跟随速度*速度百分比

3. 指令格式：json格式字符串

a. 单位：米和弧度

b. 值：**下发增量值**

c. 多模型下发二维数组 pe: 6维 pq:7维

✓ 指令格式示例

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```
// pe 和 pq按实际点位类型选择
nlohmann::json j;
std::vector<std::vector<double>> pes,pqs;
std::vector<double> pe{ 0.10,0,0,0,0,0 };
std::vector<double> pq{ 0,0,0,0,0,0,1 };
pes.push_back(pe);
j["type"] = "321"; // 支持 321 123 313
j["pe"] = pes;
j["pq"] = pqs;
```

关节

1. 跟随方式：跟随最新的位置

2. 跟随速度：

a. 最大跟随速度：关节最大速度的10%

b. 调速：最大跟随速度*速度百分比

3. 指令格式：jp --motor_pos=[0,-5,90,10,-100,0]

a. 单位：度和毫米

b. 值：**关节绝对位置**

其他辅助指令

1. 登录登出

✓

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```
// 登录
login --user=Engineer --pwd=000000
// 登出
logout
```

2. 使能

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虚拟模式：使能： `manual_en` 掉使能： `manual_ds`
真机模式：使能： `en` 掉使能： `ds`

3. 调速指令

[复制代码](#)

```
// 设置自动模式速度百分比  
set_pgm_vel --vel_percent=60  
// 设置手动模式速度百分比  
set_jog_vel --vel_percent=60
```